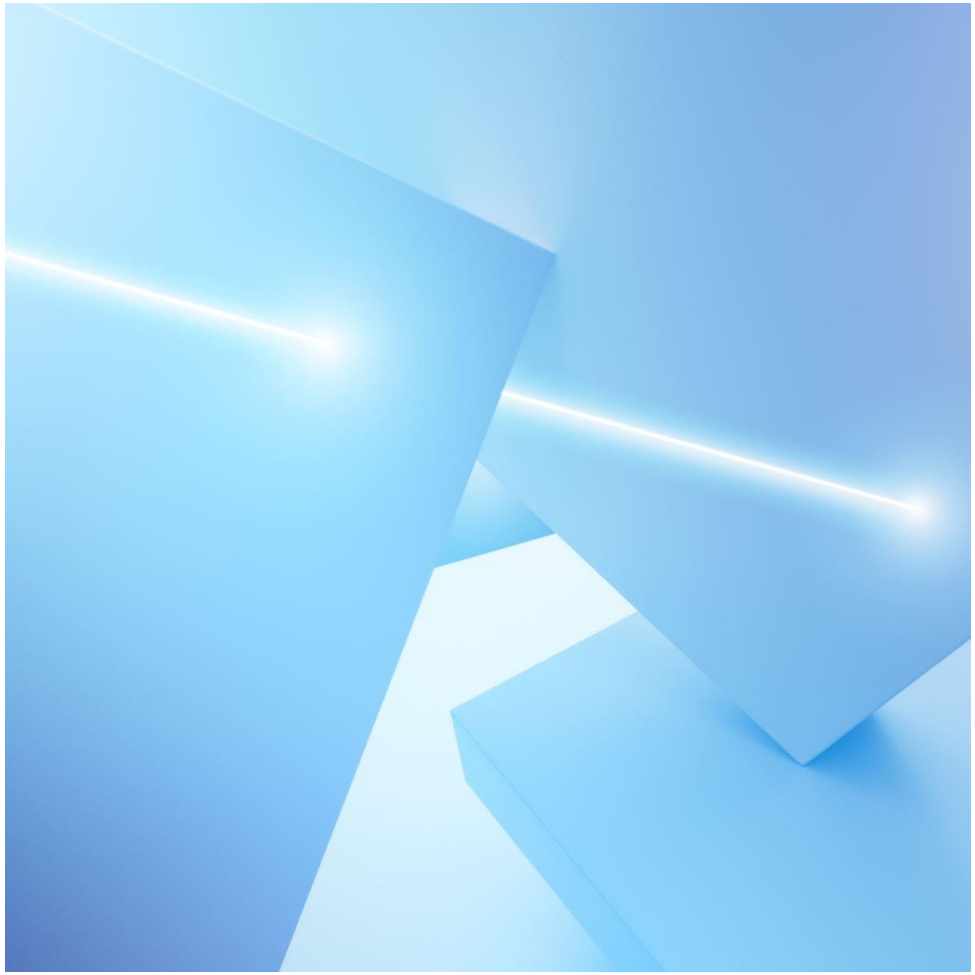
MSIX TECHNOLOGY & ENTERPRISE MANAGEMENT

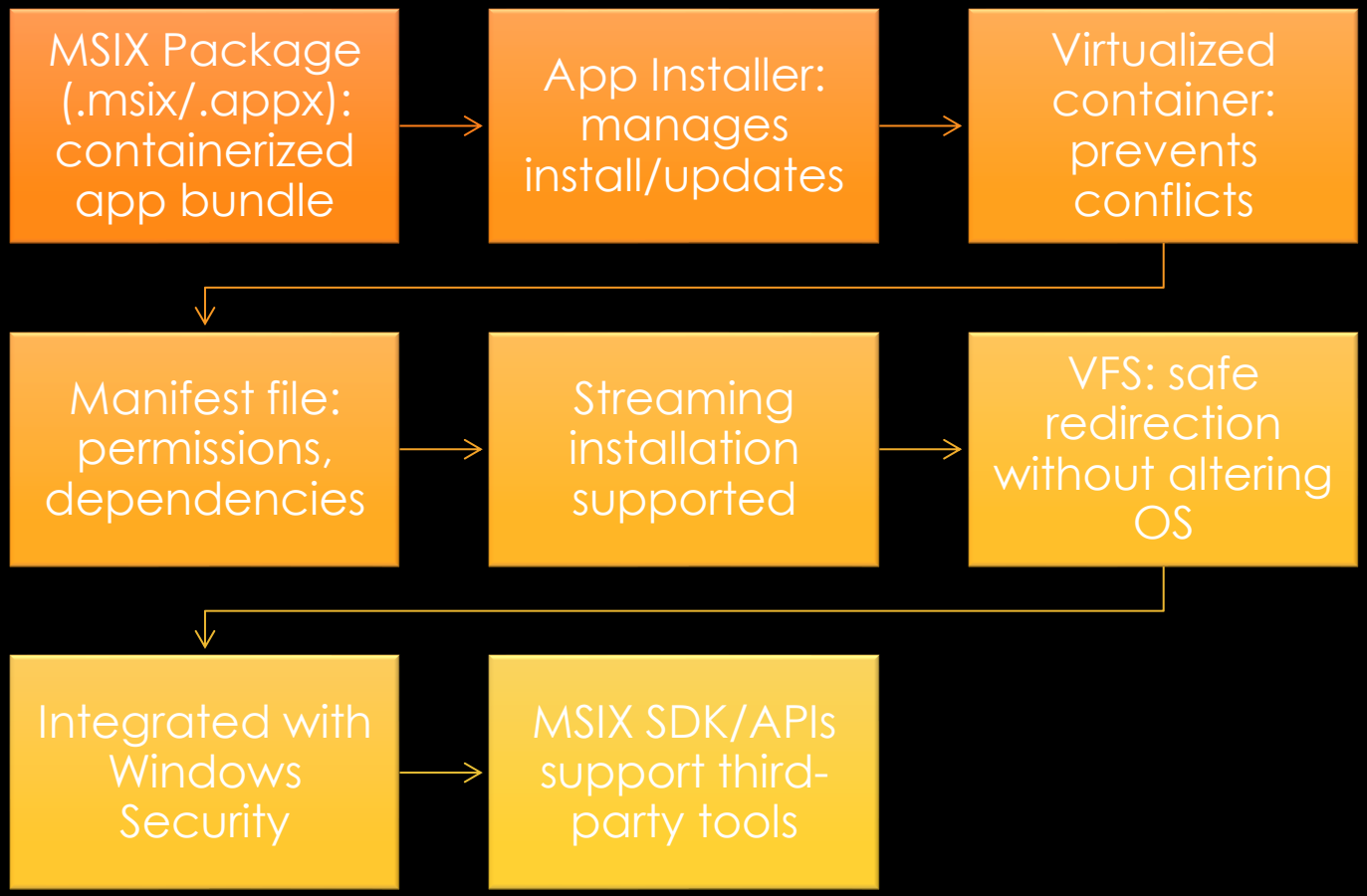
Batch A

INTRODUCTION TO MSIX

- Modern packaging format by Microsoft
- Successor of MSI, AppX, EXE
- Introduced 2018 for Windows 10/11
- Unified packaging for desktop + UWP
- Cross-platform support via MSIX SDK
- Ensures security, lifecycle management
- Minimal OS dependency
- Supported by Intune, SCCM, Azure, Store



MSIX ARCHITECTURE



MSIX WORKFLOW

1. Package creation (convert EXE/MSI)
2. Digital Signing with certificates
3. Distribution: Store, Intune, SCCM, sideload
4. Installation in isolated container
5. Update: delta updates minimize bandwidth
6. Execution without DLL conflicts
7. Rollback/repair supported
8. 100% clean uninstall

MSIX BENEFITS

- Secure: signed packages
- Clean uninstall (no leftovers)
- Storage efficiency via shared files
- Differential updates reduce downloads

- Enterprise-ready automation
- Backward compatibility (Win32, UWP)
- Self-healing apps
- Analytics for compliance

WHY MSIX?

- Legacy MSI/EXE issues: DLL Hell, leftovers
- AppX limited to UWP apps
- MSIX unifies deployment
- Supports both modern + legacy apps
- Future-proof packaging
- Enterprise benefits: security, consistency, automation

MSIX TECHNICAL LIMITATIONS

- No support for kernel drivers
- No Windows services packaging
- Compatibility issues with legacy apps
- Restricted background processes
- Streaming installs limited for big apps
- Not for non-Windows platforms
- Apps with deep OS hooks face issues

MSIX ENTERPRISE LIMITATIONS

- Requires Windows 10/11
- Complex app conversion needs testing
- Training required for IT admins
- Slow adoption due to legacy vendors
- MSIX tool lacks enterprise features
- Migration costly and time-consuming
- Sometimes dual deployment needed

MSIX PACKAGING TOOL

- Free from Microsoft Store
- Repackages legacy apps
- Convert EXE/MSI → MSIX
- Capture installation process
- Edit manifest (identity, dependencies)
- Diagnostics/troubleshooting
- Easy GUI, VS integration
- Limitations vs commercial tools

OTHER EDITORS & TOOLS

- Advanced Installer: enterprise automation
- RayPack Studio: professional repackaging
- Flexera AdminStudio: enterprise packaging/testing
- Orca: lightweight MSI/MSIX editing
- VS Extensions: developer-focused
- Free vs commercial tool comparison

SCCM OVERVIEW

- Part of Microsoft Endpoint Manager
- Centralized management
- Supports multiple OS platforms
- Application, update, OS, security management
- Used in large IT enterprises
- Key goal: secure app/update delivery

SCCM CORE FEATURES

- Software deployment automation
- Patch management via WSUS
- OS deployment (task sequences)
- Inventory collection (hardware/software)
- Compliance enforcement
- Remote control for troubleshooting
- Hybrid cloud integration with Intune

SCCM INTEGRATION

- Integrates with AD for discovery
- Azure AD + Intune support
- Endpoint protection management
- Windows Update for Business integration
- SQL Reporting dashboards
- Highly scalable for large enterprises
- Supports zero-trust compliance

PLANNING CONSIDERATIONS

- Server sizing (CPU, RAM, disk)
- SQL database requirements
- Network bandwidth planning
- AD integration for authentication
- PKI for HTTPS clients
- Site hierarchy planning
- Scalability and growth considerations

DEPLOYMENT STEPS

PREPARE WINDOWS SERVER ROLES (IIS, WSUS, .NET)

INSTALL SQL SERVER + REPORTING

LAUNCH SCCM SETUP (PRIMARY SITE)

CONFIGURE SITE CODE, DB, PORTS

SET HTTPS/HTTP COMMUNICATION

VALIDATE PREREQUISITES

MONITOR INSTALLATION LOGS

POST DEPLOYMENT TASKS

- Configure boundaries and groups
- Enable discovery methods (AD, system, user)
- Set client push installation
- Configure Software Update Point
- Create device/user collections
- Deploy test applications
- Enable alerts & compliance policies

MANAGING CAPABILITIES

- **1. Software Deployment**
- Automate distribution and installation of applications
- **2. Operating System Deployment (OSD)**
- Deploy Windows images using task sequences
- **3. Patch Management**
- Integrate with WSUS for automated patching
- **4. Inventory Management**
- Collect detailed hardware and software inventory dat
- **5. Compliance Enforcement**
- Define configuration baselines and desired settings
- **6. Remote Management**
- Remotely troubleshoot and support clients
- **7. Reporting and Analytics**
- Generate detailed reports on deployments, compliance, and inventory
- **8. Hybrid Cloud Integration**
- Co-manage devices with Microsoft Intune

COLLECTIONS & POLICIES

Collections

- Groups of devices or users managed together
- **Device Collections** (computers)
- **User Collections** (user accounts)
- **Purpose:**
- Target deployments (software, updates)
- Organize resources for reporting and management

- **Policies**

- Rules and configurations applied to clients
- **Examples:**
- Software deployment policies
- Compliance settings
- Client agent settings
- **Function:**
- Ensure clients behave consistently and meet organizational standards

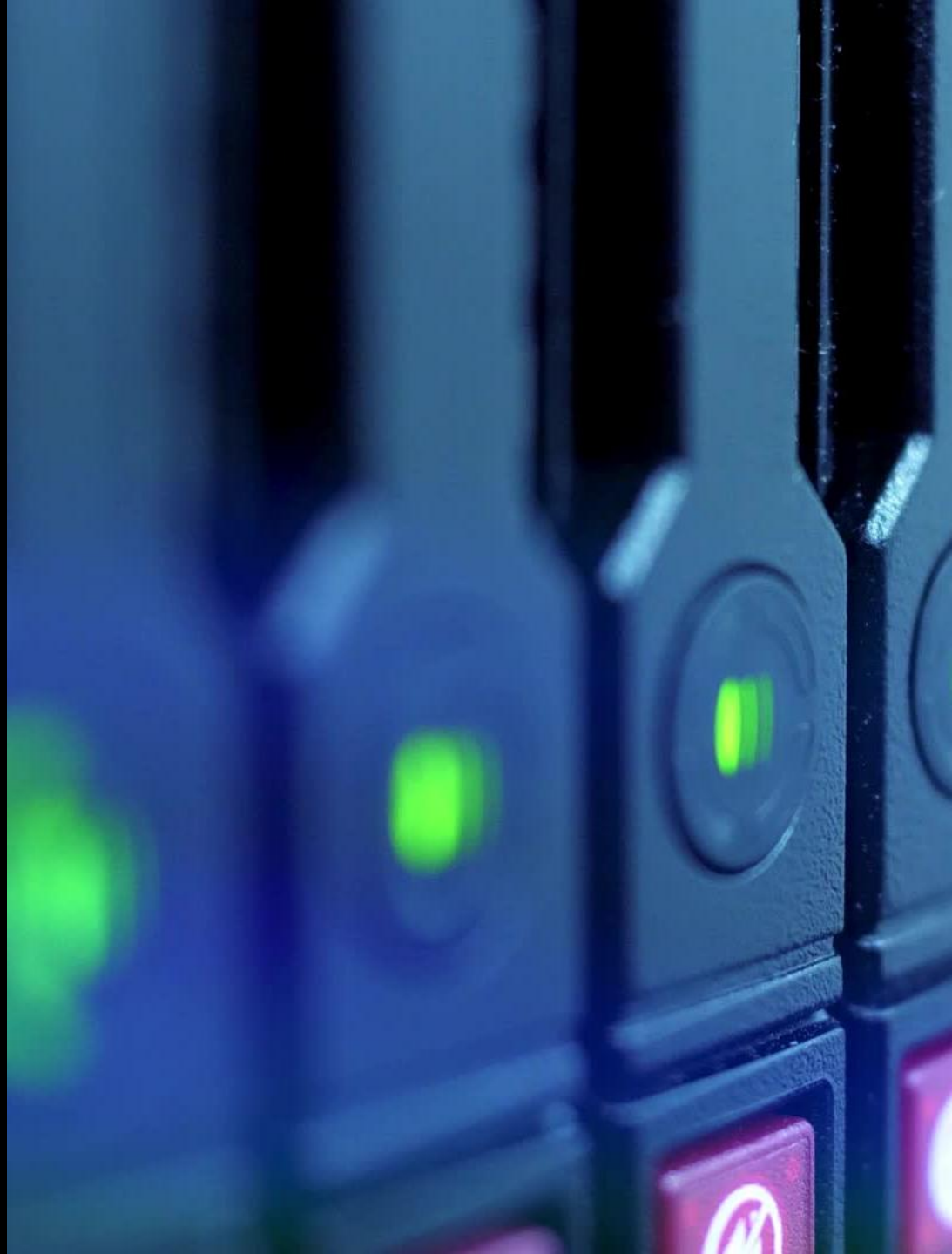
INFRASTRUCTURE REQUIREMENTS

In **SCCM (System Center Configuration Manager)**, *infrastructure* refers to the **core components and systems** required to install, run, and manage the SCCM environment effectively.

- **Site Servers** (Primary, Secondary, CAS if needed)
- **Site System Roles** (MP, DP, SUP, etc.)
- **SQL Server** (for SCCM database)

Technical prerequisites that must be in place before and during SCCM deployment to ensure successful operation.

- **Client devices** (managed by SCCM)
- **PKI (Public Key Infrastructure)**

SUPPORT & SCALABILITY

- Automated patch cycles
- User self-service portal
- Hybrid cloud readiness
- High availability with multiple servers
- Zero-trust enforcement
- Future proofing (IoT, Windows 365)
- Continuous monitoring & alerts